Background subtraction for diffraction frames: state of the problem

Subproject **pod/bgsub** (GitHub mit-eremin/pod-subtract, main at 1e10212), Dmitry Eremin, Nelson Lab, Caltech. Split out of pod/background on 2026-11-05 at background commit f1e93f5; package **bgsub**, 47 tests pass, self-contained (vendored count law, detector, masks, centre, provenance; a frozen simulator-v1 frame bank, 716 frames reproduced bit for bit). This page states the problem and where the algorithm stands; page 2 gives the measured performance, the known defects and the next steps. Session ids are anonymous (sids).

1. The problem, in the owner's words (re-scope 2026-11-04, scope correction 2026-11-05)

A very **fast**, **mathematically correct**, **universal** background estimation and subtraction algorithm for electron-diffraction frames, better than the flat local backgrounds of XDS / DIALS (flat boxes are biased on a curved background and lose intensity; their code is not copied). Correct means: unbiased background **everywhere** and **under masked spots**; honest per-pixel background **variance**; right behaviour under the detector's real **count statistics**; right **near the beam stop, grid bars and crystals**; and a fast estimate (target <= 50 ms per 1024^2 frame).

- **Background only.** Peak integration is not the product and the result is not for XDS. The integrators in the code (G-ML, LC, intervals, coverage) were the orchestrator's addition; they stay but nothing is judged on them. Injected peaks remain a *test*, scored against an oracle-background subtraction, never against true intensity.
- **No other packages** (no DIALS, nXDS); methods are developed here. The physics forward model / simulator stays in pod/background and is a test bench only.
- **Universal** today means Apollo (bin 4, 8, 16) with its stored-count law; CETA-D and TVIPS need their own variance models before any claim.

2. The approach (v1 default; v2, v2.1 opt-in; v0 reproducible)

The frame's stored counts are reduced to sufficient statistics on polar cells (1-px rings x 64 sectors x 2 checkerboard halves) and fitted with a **linear smooth model in stored counts**: a fine radial cubic B-spline (rings, steep halo) + azimuthal harmonics m <= 6 on a coarse radial spline + centre / ellipticity derivative columns + (v1) penumbra columns on the 1-4 px next to the beam stop; (v2) re-centring and a shrunken 1-px ring refinement, (v2.1) on elliptical rings. The fit is **cross-fitted weighted least squares**: the weights are the count-law variances taken from the *other* checkerboard half, so the estimator is linear in the counts and unbiased for *any* count law whenever the truth lies in the model's span (no median, no Poisson assumption). Spots, hot pixels and zingers are excluded by one-sided count-law tests (alpha 10^{-6} per pixel and per 3 x 3 window); spots the caller will measure are always excluded by discs. Variance is a sandwich with the frame's Pearson dispersion. The hybrid **H2.1** adds a local annulus residual only where its signal is significant (T > 2 with block standard errors), with the spot's own light removed (spill correction); otherwise it is the global **G2.1**.

The Apollo count law it carries: stored value = round-half-even of the 3-raw-frame electron sum / m (m = 1, 4, 16 e per count at bin 4, 8, 16), tie rule p = 0.403, paralyzable dead time tau = 0.206 per e per physical pixel per raw frame; archive TIFFs are MRC + 1; seams and borders masked.

3. Where it stands: v2.1 against its own pre-registered rules (2026-11-05)

Every version is pre-registered (rules committed before the test run, no parameter changes after it, failures reported with their cells). The verdict rule: VALID iff R1, R2, R3 and R5 pass and R4 passes in the F and classic regimes. **v2.1 is NOT VALID, by two narrow margins, both about the background.**

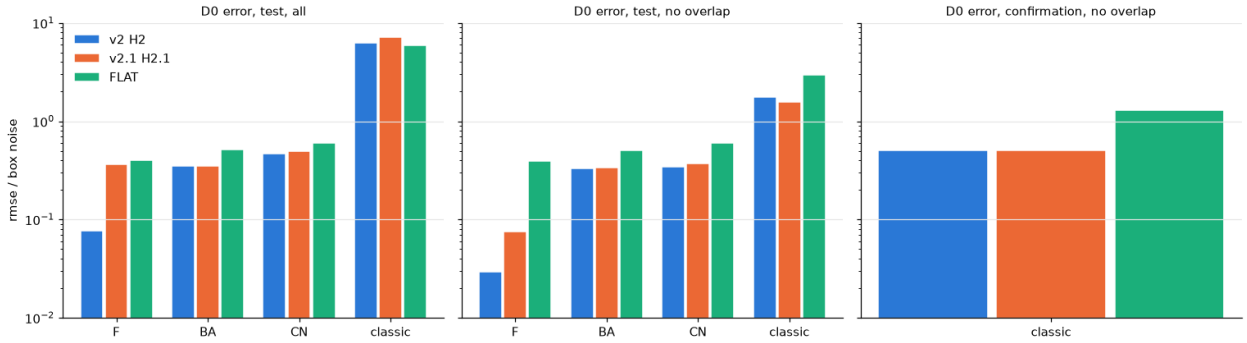
Rule	What it tests	v2 (previous)	v2.1 (current)	v2.1 status
R1	Synthetic unbiasedness: 60 lacey cells (bin 4 / 8 x dose x1 / 3 / 10 x plain / structured x D0-D4), |beta| <= max(0.5 %, 3.2 SE), for G and H	G pass 0/60; H fail 2/60 (arm cell +0.70 %, sparse cell -3.16 %)	G2.1 pass 0/60; H2.1 fail 1/60: sparse bin-8 cell (dose x3, plain, D1; 0.2 stored counts per box) -2.46 +- 0.72 %; arm cell fixed (+0.34 %)	FAIL 1/60 (H); no power at 0.2 counts/box
R2	Count statistics / truncation: spot-free twins, 24 cells, paired bias of G <= max(0.3 %, 3.1 SE); masking on - off <= 0.1 %	pass (max 0.45 %)	pass (max 0.45 %)	PASS
R3	Sigma honesty and interval coverage of the injected-spot estimates (integration; out of scope now)	pass; coverage 0.57-0.58	pass; coverage 0.66-0.69 (fix 3)	PASS (not a product criterion)
R4	Real frames, better than FLAT per regime x d bin, spots without host overlap: F, BA, CN, classic (17 fly-scan + 20 classic test sessions, 48,930 spots; 9 confirmation sessions)	P / P / P / P	P / P / P / P; confirmation P; all-spot classic F (host Bragg overlap, not background)	PASS
R5	Speed of the background estimate: 1024^2, numba, plan cached, <= 50 ms on this Mac	pass 46.8 ms (own run), 50.2 ms in the v2.1 run	54.2 ms under load (v1 46.3 ms in the same run); elliptical ring kernels +2.7 ms (A278)	FAIL 54.2 ms

R6 and R7 (sharp ring) are reported, not in the verdict. Report of record: notes/reports/2026-11-05_bgsub_algorithm_v2_1.md (Sec. 2.4 rules, Sec. 7 verdict, Sec. 8 limits).

4. Measured performance of the background estimate

Background error next to the beam stop (D0, d 4-8 A) on real Apollo frames with injected peaks, in units of the spot box's own count noise, spots without a host overlap. Smaller is better; FLAT is the XDS / DIALS-style local flat background on the same spots. Synthetic plain lacey frames: G2.1 and H2.1 0.07-0.16 vs FLAT 0.38-0.62; synthetic structured frames at D4: H2.1 0.39-0.40 vs v2 0.45-0.49 vs G 0.84-0.97.

Regime (real frames)	Spots	v2.1 H2.1 (hybrid)	v2.1 G2.1 (global)	v2 H2	FLAT
F: fly-scan, clean film, bin 8	13,650	**0.08**	0.32	0.03	0.40
BA: bar-adjacent, bin 8	14,280	**0.34**	0.46	0.33	0.50
CN: crystal neighbour, bin 8	12,600	**0.37**	0.57	0.35	0.60
classic, bin 4 (20 test sessions)	8,400	**1.57**	1.58	1.76	2.98
classic confirmation (9 sessions)	3,430	0.50	**0.48**	0.51	1.30



D0 background error (rmse / box noise, log scale), v2 H2 vs v2.1 H2.1 vs FLAT: left all test spots, middle test spots without a host overlap (the verdict form), right the classic confirmation set. With all spots, classic H2.1 7.20 vs FLAT 5.96 and F 0.36: sample Bragg spots of the host under the injected spot (an overlap, not background; A266, A279). Outside D0, H2.1 = G2.1 within 0.01-0.03. Figure: notes/figures/bgsub_v2_1/real_d0_v2_v21.png.

5. Known defects and limits

- **Sparse counts (R1).** At 0.2 stored counts per box the gate fires at its nominal 2.5 % rate on noise and five firings move a cell by 0.5 %; the failing cell also has G2.1 itself at 3.05 SE on those frames (on 64 fresh frames H2.1 is -0.47 +- 0.72 %, A276). The test had no power; it needs a re-test with enough frames and possibly a count-level-aware gate.
- **Sharp rings (A277, R7 reported).** A film ring narrower than the spline knots (classic sb7857cd8, 4.6 A) is not removed (ring-zone dispersion 16.2 -> 15.1 against a target of 8.1; no harm elsewhere, max ratio 1.002): 34-81 % of its pixels are count-law flags and the skipped ring mean breaks down. A full-frame robust ring term was tried and rejected (it breaks the background next to the stop, A275); a ring-zone-only high-breakdown first pass is the open option.
- **Speed (A278).** 54.2 ms at 1024^2 against 50 ms, measured under load; the elliptical kernels cost +2.7 ms and can be fused into existing passes (~2.5 ms back).
- **Structure outside the model.** Anything neither smooth on the model's scales nor linear at the annulus scale is not modelled; a compact host overlap under a spot is invisible to the probes (A279).
- **Variance at bin 16** is not honest (saturated cores on the wrong dead-time branch, A196); bin 16 is synthetic-only. The penumbra columns are ignored by var_map today.
- **Detector coverage.** Apollo bin 8 (fly-scan) and bin 4 (classic) are tested on real frames; **CETA-D bin 2 and TVIPS F30 are untested** and need an integrating-detector variance model (gain per electron, read noise, offset, no rounding) before any run. Evaluation-code defect A281 (duplicated rows in the v2 / v2.1 pull tables; R4 and D0 unaffected) is open.

6. Next steps (ROADMAP.md, in order) and what the owner must decide

- **1. Background-only validity spec**, pre-registered: bias everywhere and under masked spots with stated power per cell; per-pixel variance honesty (replicates on synthetic frames, split-half / twins on real); zones (stop, penumbra, arm, bar edges, crystal neighbours); count-law regimes (bin 4 / 8 / 16, sparse to bright, gain on / off, dead-time ceiling, zero level); speed on an idle machine; injected peaks only against an oracle background. Replaces the integration-flavoured R1-R7.
- **2. Bring v2.1 to valid under it:** fuse the elliptical kernels (< 50 ms); a power-adequate sparse-cell re-test; fix A281 when the real-frame drivers are ported from legacy/podbg_scripts; optionally the ring-zone first pass.
- **3-5.** Integrating-detector variance model, then CETA-D bin 2 validation; TVIPS F30 validation (ringcal's readers, vendored); honest variance at bin 16.
- **Owner decisions (ask, do not decide):** (a) **output format** per frame: proposed a float32 background map + per-pixel variance map in stored counts, the subtracted frame on request, and a sidecar (version, count law, zero level, mask, centre); alternatives are radial-model coefficients + covariance, or electrons; decide before any file writer or CLI is built. (b) **datasets** for CETA-D bin 2 and TVIPS F30 (which sids; whether dark / flat frames exist).